Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

PH506 – ATOMIC PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Give any two properties of cathode rays.
2. Mention any two advantages of Aston's mass spectrograph.
3. What is Bohr magneton?
4. What do you mean by LS coupling?
5. What is Zeeman effect?
6. State Larmor's theorem.
7. Write down Einstein' photoelectric equation.
8. Provide any two applications of photoelectric cell?
9. Give any two properties of X-rays.
10. State Bragg's law.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Write a note on magnetron method for the specific charge of an electron.

(or)

b) How will you determine e/m of positive ions by using Dempster's mass spectrograph.

12. a) Explain magnetic dipole moment due to orbital motion of the electron.

(or)

b) Give the theory of Stern-Gerlach experiment.

13. a) Explain Paschen Back effect.

(or)

b) State and explain Stark effect.

14. a) Describe Lenard's method to determine e/m for photo electrons.

(or)

b) Write a note on photovoltaic cell.

15. a) Explain the characteristics of X-ray spectrum.

(or)

b) Write a note on Laue method of X-ray diffraction.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Describe with theory Thomson's parabola method for determining e/m of positive ions.

17. a) Explain vector atom model.
    b) Discuss the quantum numbers associated with the vector atom model.

18. Describe the experimental determination of critical potentials by Davis and Goucher's method.

19. Describe Richardson and Compton experiment.

20. a) What is a reciprocal lattice?
    b) With a diagram explain powder X-ray diffraction method.

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